

A constraint programming approach for resource-constrained flexible assembly flow shop scheduling problem with batch direct delivery

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ABSTRACT

Production and distribution are both crucial components of supply chains. Integrated production and distribution scheduling (IPDS) in the context of flexible assembly flow shop scheduling and batch delivery problems is often overlooked. A realistic problem inspired by the production and distribution processes of a dishwasher factory can be modeled as a resource-constrained flexible assembly flow shop scheduling problem with batch direct delivery (RCFAFSP-BDD). In this problem, order requirements are decomposed into several production tasks processed at different stages of the workshop, and are then delivered in batches via a third-party logistics provider to regional distributors at various locations. Auxiliary resource restrictions, hierarchical coupling constraints, machine eligibility restrictions and sequence-dependent setup times, are incorporated into the problem as operational constraints. To the best of our knowledge, this is the first attempt to solve this problem. This work formulates a mixed-integer linear programming (MIP) model to minimize total costs, including tardiness, inventory, and delivery costs. Given the problem's strong NP-hard nature, the focus is on developing an efficient solution approach using constraint programming (CP). A CP model is proposed and enhanced with multiple redundant constraints. To reduce runtime, two branching strategies are designed for the CP model. Numerical experiments with varying instance scales reveal that the proposed CP model outperforms the MIP model in accuracy and efficiency within the given time limit. The redundant constraints and search strategy can reduce CP model runtime by up to 263.83%. Compared to manual scheduling at the studied factory, the CP model can cut costs by up to 26.59% for real data, offering viable alternatives for factory planners.

1. Introduction

Integrated production and distribution scheduling (IPDS) is a promising approach in industrial manufacturing for reducing costs and enhancing efficiency from a supply chain perspective. As an extension of traditional scheduling problems, IPDS involves not only determining the optimal delivery batches for completed orders but also making scheduling decisions. One key focus is to enable parallel production and packaged delivery of multiple orders to the same address by coordinating schedules within the production process. Existing research on IPDS can be categorized by delivery characteristics into individual and immediate delivery, splittable delivery, shipping with fixed delivery departure dates, and batch delivery methods (Chen, 2010). Batch delivery is regarded as the most effective strategy for reducing delivery costs by minimizing the total number of delivery batches (Bachtenkirch and Bock, 2022). It can be further divided into two modes: batch delivery by direct shipping, where only orders to the same address are delivered together in a single shipment, and

batch delivery with routing (Chen, 2010). The former is commonly used in production-distribution supply chains between home appliance manufacturers and regional distributors. Manufacturers depend on third-party logistics providers to handle point-to-point deliveries to each distributor, while distributors manage regional distribution and circulation. Many tiny orders from distributors can accumulate significant demand for various products, creating practical opportunities for multi-batch, point-to-point distribution tasks (Noroozi et al., 2018).

Existing research on IPDS involving batch direct delivery primarily addresses single machines (Fu et al., 2018), parallel machines (Zhang et al., 2022d), flow shops (Qin et al., 2021), and distributed shops (Yang and Xu, 2021). However, there has been limited investigation into the assembly scheduling problem (ASP), which includes both processing and assembly stages. These environments must not only coordinate the scheduling of orders for production and batch delivery but also manage the production schedules of related components with strict hierarchical relationships. This hierarchical coupling (Zou et al., 2018), arising from

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the product's assembly structure, presents significant challenges for production scheduling. Current research on the ASP can be summarized as follows: (1) Most studies focus on the two-stage assembly flow shop scheduling problem (AFSP) (Komaki et al., 2019). In reality, products often have more complex structures that involve multiple production stages within the workshop system (Shi et al., 2020). (2) Research typically considers only machines as resources. However, in practice, production also requires the efficient planning of auxiliary resources such as molds, fixtures, and skilled workers. (3) The ASP is usually addressed using well-designed heuristic or meta-heuristic algorithms in existing research.

This article aims to address the resource-constrained flexible assembly flow shop scheduling problem with batch direct delivery (RCFAFSP-BDD). In this problem, orders from multiple regional distributors must be processed through a flexible assembly flow shop with three production stages, and appropriate batches are allocated for delivering finished orders to various locations. The problem also takes into account constraints on auxiliary resources. The objective is to minimize the total cost, which includes tardiness, inventory, and delivery expenses. To the best of our knowledge, this specific problem has not been explored previously. The two-stage AFSP with batch delivery has been shown to be NP-hard (Basir et al., 2018). The three-stage RCFAFSP-BDD extends this complexity by introducing additional constraints, maintaining its NP-hard status. To address this problem, we propose both a mixed-integer linear programming (MIP) model and a constraint programming (CP) model. We also introduce several redundant constraints and search strategies to improve the accuracy and efficiency of the CP model. The performance of these models and strategies is evaluated and compared through extensive numerical experiments. The main novelties and contributions of this article are as follows:

- A novel three-stage RCFAFSP-BDD system is introduced and investigated for the integrated scheduling decision of production and delivery of orders, filling the gap in this field.
- A MIP mathematical model is designed to formulate RCFAFSP-BDD.
- A new approach based on the CP is proposed for considered problem.
- Several redundant constraints and search strategies have been designed to further improve the solution quality and efficiency of the CP model.

The rest of this article is organized as follows. Section 2 provides a literature review, Section 3 details the problem with an example, Section 4 discusses the CP formulation, redundant constraints, and search strategies, Section 5 presents extensive numerical experiments, and Section 6 offers the final conclusions.

2. Literature review

Since this article addresses the flexible assembly flow shop scheduling problem (FAFSP) with various practical features, the literature review is divided into three subsections. First, it examines studies on the major variants of the FAFSP. Next, it explores the FAFSP with batch direct delivery. Following that, it investigates contributions related to the interaction between auxiliary resources and the FAFSP. Finally, to highlight the novelty of this study, a comparison with existing works is provided, emphasizing the common factors.

2.1. Flexible assembly flow shop scheduling problem

According to Framinan et al. (2019), AFSP, a prominent type of assembly scheduling problem, is defined as $\alpha 1 \rightarrow \alpha 2 | \beta | \gamma$, where $\alpha 1$ and $\alpha 2$ represent the machine layouts for the processing and assembly stages, β denotes special constraints, and γ specifies the optimization objectives. Existing AFSP research is typically categorized into two types: two-stage AFSPs and multi-stage AFSPs. Most studies focus on two-stage

AFSPs (Lei et al., 2021; Zhang and Tang, 2022; Talens et al., 2022), while FAFSP represents a multi-stage AFSP, involving configurations like $DPm \rightarrow Fm$, $DPm \rightarrow HFm$, and $DFm \rightarrow 1$. Here, DPm refers to dedicated parallel machines, DFm to flow shop layouts in the processing stage, and Fm and HFm to flow shop and flexible flow shop layouts in the assembly stages, respectively.

Research on the $DPm \rightarrow Fm$ problem, particularly $DPm \rightarrow F2$ (Zhang et al., 2021; Fernandez-Viagas et al., 2022), has produced several results. However, $DPm \rightarrow HFm$, a more complex extension, still sees limited research (Framinan et al., 2019). The three-stage $DPm \rightarrow HFm$ problem is the most common variant and has been investigated by Xiong et al. (2015), who analyzed a three-stage FAFSP involving multiple parallel processing machines, an assembly machine, and dedicated machines. They developed a MIP model aimed at minimizing total flow time and proposed several efficient algorithms for solving it. Following this, Zhang et al. (2022a) introduced a novel distributed co-evolutionary memetic algorithm for more efficient solutions. They also extended the problem by incorporating limited buffer constraints and proposed a dual-space cooperative co-evolutionary memetic algorithm to address these constraints (Zhang et al., 2022c). Further extending this research, Zhang et al. (2022b) adapted the three-stage FAFSP to distributed manufacturing environments, proposing an elite archive-assisted adaptive memetic algorithm. Recently, Yuan et al. (2023) extended the problem to include machine maintenance, assuming a fixed maintenance period during which the first-stage machine cannot process jobs. They focused on minimizing completion time and developed three approximation algorithms for this purpose. Additionally, Li et al. (2024) examined the FAFSP considering machine deterioration and proposed a discrete artificial bee colony algorithm, which demonstrated superior performance compared to existing algorithms in their tests.

Studies on manufacturing systems with more than three stages, such as those extending $DFm \rightarrow 1$ (Shao et al., 2020; Hamzadayi et al., 2021), are indeed insightful. For instance, Lei and Zheng (2017) investigated a multi-stage FAFSP with a flexible flow shop layout for the processing stages and a single assembly machine. Their objective was to minimize total tardiness, maximum tardiness, and makespan simultaneously. They developed a neighborhood search algorithm to tackle this complex problem.

In summary, while there is a limited body of research on FAFSP, recent studies have shifted toward more realistic simulations of manufacturing scenarios. This includes integrating practical constraints and combining FAFSP with other decision problems. Such developments highlight the ongoing evolution and growing sophistication in this research area.

2.2. Scheduling with batch delivery

Numerous efforts have focused on IPDS with batch direct delivery (Noroozi et al., 2018; Li et al., 2021; Azad et al., 2022; Zhang et al., 2022d). Noroozi et al. (2018) first explored this within the order acceptance scheduling problem (OASP), aiming to balance accepted order revenue, delivery costs, and tardiness penalties to maximize overall benefit. Due to the problem's NP-hard nature, an adaptive genetic algorithm was proposed. Li et al. (2021) examined IPDS in the context of distributed flow shop scheduling (DFSP) with batch delivery, developing a whale optimization algorithm to minimize completion time and energy consumption. Azad et al. (2022) addressed IPDS within permutation flow shop scheduling (PFSP), focusing on sequence-dependent setup times and vehicle routing, and the goal of their work is to lower the overall cost of tardiness and batch distribution in the supply chain. To solve the problem, a mixed-integer nonlinear programming model was designed and a genetic algorithm was proposed to solve the model. Zhang et al. (2022d) consider the integrated scheduling problem of production and delivery on parallel batch processing machines with non-identical capacities in different

locations in cloud manufacturing. A MIP model was designed with the optimization objective of minimizing the total service completion time, and an improved particle swarm optimization algorithm was developed for solving the problem. Recently, Hajibabaei and Behnamian (2024) studied the flexible assembly job shop scheduling problem (FAJSP) considering cleaner production criteria, interruptions, and batch delivery. The objective is to minimize the costs of tardiness, earliness, delivery, and maximum completion time, along with minimizing the amount of speed increase and carbon dioxide emissions. A hybrid algorithm consisting of NSGA-II and memetic were proposed to solve the problem.

Research on IPDS with AFSP remains relatively sparse. Wang et al. (2016) was the first to address the IPDS problem within the context of AFSP ($DPm \rightarrow 1$ type), focusing on batch delivery for a single customer. They proposed two efficient heuristics and a new hybrid meta-heuristic algorithm to tackle the problem. Kazemi et al. (2017) expanded this work by considering the IPDS problem in an AFSP ($DPm \rightarrow Pm$ type), where Pm refers to parallel assembly machines. They addressed batch delivery demands for multiple customers, developed a model aimed at minimizing the sum of tardiness and delivery costs, and applied a hybrid imperialist competitive algorithm for solution. Basir et al. (2018) further investigated IPDS problems in the context of AFSP ($DPm \rightarrow Pm$ type), incorporating vehicle capacity constraints. They used an improved genetic algorithm with a layered decision-making strategy to solve the problem. Recently, Yang and Xu (2021) examined IPDS problems within the framework of the distributed assembly permutation flow shop problem (DAFPSP/ $DFm \rightarrow 1$), considering the delivery demands of multiple customers. They constructed a mathematical model to minimize tardiness and delivery costs and designed various heuristics and local search algorithms for efficient problem-solving.

In conclusion, the current literature on IPDS primarily addresses contexts such as OASP, PFSP, DFSP, and two-stage AFSP. Research into IPDS within the framework of three-stage FAFSP ($DPm \rightarrow HFm$) remains unexplored. However, in real-world appliance manufacturing, the complexity of product assembly structures often leads to more production stages, highlighting a need for further investigation in this area.

2.3. Scheduling with auxiliary resources constraint

Significant efforts have been directed toward various scheduling problems, including the parallel machines scheduling problem (PMSP) (Zheng and Wang, 2018), the flexible job shop scheduling problem (FJSP) (He et al., 2023), the flexible flow shop scheduling problem (FFSP) (Hasani and Hosseini, 2022), and the distributed flexible flow shop scheduling problem (DFFSP) (Tao et al., 2022). Among these, the resource-constrained parallel machine scheduling problem (RCPMSP) has received extensive research (Afzalirad and Shafipour, 2018; Vallada et al., 2019). For example, Zheng and Wang (2018) tackled a resource-constrained unrelated parallel machine scheduling problem (RCUPMSP) with green manufacturing considerations, focusing on minimizing both the maximum completion time and total carbon emissions using a collaborative multi-objective fruit fly optimization algorithm. Afzalirad and Shafipour (2018) explored a RCPMSP with machine eligibility constraints, proposing an integer programming (IP) model and a genetic algorithm to minimize completion time. Yunusoglu and Yildiz (2022) addressed an RCUPMSP incorporating auxiliary resource constraints, sequence-dependent setup times, precedence relations, machine eligibility constraints, and release dates, aiming to maximize completion time with a CP model and two branching strategies to reduce computation time. Vallada et al. (2019) studied an RCUPMSP with a focus on minimizing the maximum completion time, introducing several meta-heuristics, including a scatter search algorithm, for an efficient solution. Meanwhile, Hasani and Hosseini (2022) investigated the resource-constrained flexible flow shop scheduling problem (RCFFSP), considering machine eligibility constraints and sequence-dependent setup times, and in the first stage, parallel machines were

unrelated and auxiliary resources were required to process the jobs. They developed a MIP model and genetic algorithm to minimize completion time. He et al. (2023) addressed a resource-constrained flexible job shop scheduling problem (RCFJSP) with machine and worker constraints, employing an African vulture optimization algorithm to minimize maximum completion time and total delay. In a recent study, Tao et al. (2022) considered a distributed resource-constrained hybrid flow shop scheduling problem (DRCHFSP) and developed an adaptive artificial bee colony algorithm to minimize the maximum completion time. Additionally, Liu et al. (2024) explored an RCFJSP with fixture-pallet combinatorial optimization, creating a MIP model and a novel genetic algorithm with feasibility correction and self-learning variable neighborhood search to minimize makespan. Fekri et al. (2024) examined an RCFJSP requiring additional human resources, designing a MIP model to minimize total completion time and idle time, and developed genetic and simulated annealing algorithms for solutions. Finally, Barak et al. (2024) addressed an RCFJSP with operator allocation, aiming to minimize operator idleness cost, total tardiness, and earliness penalties using a multi-objective invasive weed optimization algorithm.

Currently, research incorporating auxiliary resource constraints into the AFSP remains limited. Hosseini et al. (2022) were the first to explore an AFSP constrained by both the number of auxiliary resources and machine eligibility in the first stage, while also accounting for sequence-dependent setup times and job release times. They developed a novel MIP model and introduced a new heuristic algorithm to provide efficient solutions. Building on this, Liu et al. (2022) expanded the scope by considering parallel teams and worker skills within the AFSP. They examined a complex product assembly line scheduling problem that incorporates multi-skilled worker assignments and parallel team scheduling. Their approach involved an IP model with dual optimization objectives: minimizing maximum completion time and balancing team workload. To solve this problem effectively, they proposed three improvement strategies based on multi-objective evolutionary algorithms.

In summary, current research on scheduling problems is increasingly reflecting real-world scenarios through more realistic simulations. While efforts regarding auxiliary resource constraints are advancing, particularly in the context of the PMSP, there is a notable gap in literature addressing these constraints within the FAFSP.

Table 1 provides a summary of previous contributions related to the problem of interest and highlights the novelty of this work. As illustrated in the table, there has been growing attention in recent years to two key areas: the IPDS problems and the incorporation of auxiliary resource constraints in production scheduling. However, existing literature has yet to address the FAFSP with auxiliary resource constraints during the production stage and the subsequent delivery planning for finished orders. The main contribution of this article is to bridge this gap by addressing a problem inspired by the dishwasher production process.

3. RCFAFSP-BDD statement

The RCFAFSP-PSBD problem is modeled after the production and distribution processes of a dishwasher manufacturing plant. Utilizing the concept of product modularization, the complex production operations are divided into distinct stages. Each stage's multiple operations are modularized and treated as a single operation. A simplified process diagram for dishwasher production is illustrated in Fig. 1. The production process is comprised of three stages: cutting and stamping metal sheets to form the shell, pre-assembling internal components (e.g., motors and control panels), and final assembly of the remaining electronic components into the dishwasher.

In this section, we first provide a detailed description of the problem structure, followed by the presentation of a MIP model for addressing it. Finally, a simple example is used to clarify and enhance understanding of the problem.

Table 1

Comparison contributions on the RCFAFSP-BDD.

Reference	Problem	Main features						Auxiliary resource		Number of customers		Delivery		Objective		Solution method
		TS	SDST	ME	LB	RT	PC	Eq.	EWf	1	Multi	Direct	Round trip	Time-based	Cost-based	
Xiong et al. (2015)	FAFSP													✓		MIP, HGA-VNS
Lei and Zheng (2017)	FAFSP													✓		NSG
Zhang et al. (2022a)	FAFSP													✓		DCMA ^a
Zhang et al. (2022c)	FAFSP				✓									✓		DCMA ^b
Zhang et al. (2022b)	FAFSP					✓								✓		MIP, EAMA
Fernandez-Viagas et al. (2022)	FAFSP			✓										✓		CH
Li et al. (2022)	FAFSP			✓										✓		B&B, ITS
Yuan et al. (2023)	FAFSP													✓		AA
Li et al. (2024)	FAFSP													✓		DABC
Zheng and Wang (2018)	UPMSP							✓						✓		MIP, CMFOA
Afzalirad and Shafipour (2018)	UPMSP			✓				✓						✓		IP, GA
Vallada et al. (2019)	UPMSP							✓						✓		SSA
Yunusoglu and Yildiz (2022)	UPMSP	✓		✓		✓	✓	✓						✓		CP
Hasani and Hosseini (2022)	FFSP	✓		✓					✓					✓		MIP, GA
Tao et al. (2022)	DFFSP							✓						✓		MIP, SABC
Hosseini et al. (2022)	AFSP	✓		✓		✓		✓						✓		MIP, CH
Liu et al. (2022)	AFSP								✓					✓		IP, MOEA
He et al. (2023)	FJSP			✓					✓					✓		IABOA
Liu et al. (2024)	FJSP			✓				✓						✓		MIP, IGA
Fekri et al. (2024)	FFSP			✓					✓					✓		MIP, GA, SA
Barak et al. (2024)	FJSP	✓							✓					✓	✓	MIP, MOIWO
Wang et al. (2016)	AFSP									✓		✓		✓	✓	HGA-OVNS
Kazemi et al. (2017)	AFSP										✓	✓		✓	✓	MIP, ICA
Basir et al. (2018)	AFSP										✓	✓		✓	✓	MIP, IGA
Noroozi et al. (2018)	OASP											✓		✓	✓	AGA
Yang and Xu (2021)	DAPFSP											✓		✓	✓	CH
Li et al. (2021)	DFSP											✓		✓		WOA
Azad et al. (2022)	PFSP	✓									✓		✓		✓	MIP, GA
Zhang et al. (2022d)	UPMSP			✓		✓					✓	✓		✓		MIP, IPSO
Hajibabaei and Behnamian (2024)	FAJSP					✓					✓	✓			✓	MIP, NSMA
This study	FAFSP	✓		✓				✓			✓	✓			✓	MIP, CP

Abbreviations: TS: Task splitting; SDST: Sequence-dependent setup times; ME: Machine eligibility; LB: Limited buffer; RT: Release time; PC: Precedence constraints. Eq.: Equipment; EWf: Expert workforce.

HGA-VNS: Hybrid genetic algorithm-variable neighborhood search algorithm; NSG: Neighborhood search with global exchange; DCMA^a: Distributed co-evolutionary memetic algorithm; DCMA^b: Dual-space co-evolutionary memetic algorithm; EAMA: Elite archive-assisted adaptive memetic algorithm; CH: Constructive heuristics; B&B: Branch-and-bound algorithm; ITS: Improved tabu search; AA: Approximation algorithms; DABC: Discrete artificial bee colony algorithm; CMFOA: Collaborative multi-objective fruit fly optimization algorithm; IP: Integer programming; GA: Genetic algorithm; SSA: Scatter search algorithm; SABC: Adaptive artificial bee colony algorithm; MOEA: Multiple objective evolutionary algorithm; IABOA: Improved African vulture optimization algorithm; IGA: Improved genetic algorithm; MOIWO: Multi-objective invasive weed optimization algorithm; HGA-OVNS: Hybrid meta-heuristic; ICA: imperialist competitive algorithm; AGA: Adaptive genetic algorithm; WOA: Whale optimization algorithm; IPSO: Improved particle swarm optimization algorithm; NSMA: Hybrid algorithm consisting of NSGA-II and memetic.

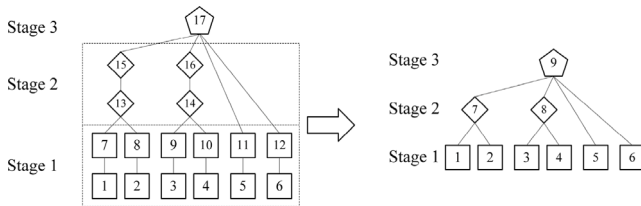


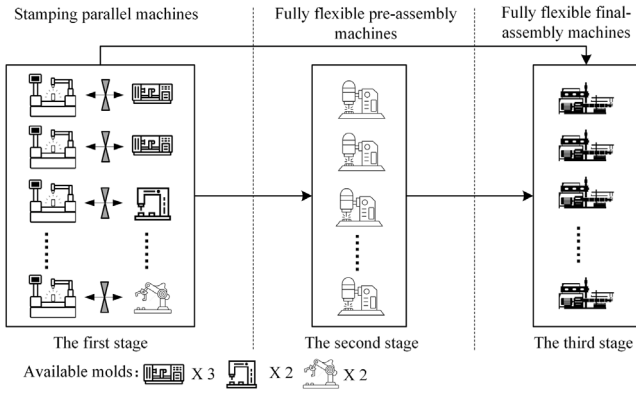
Fig. 1. Simplified diagram of home appliance product process.

3.1. Problem definition

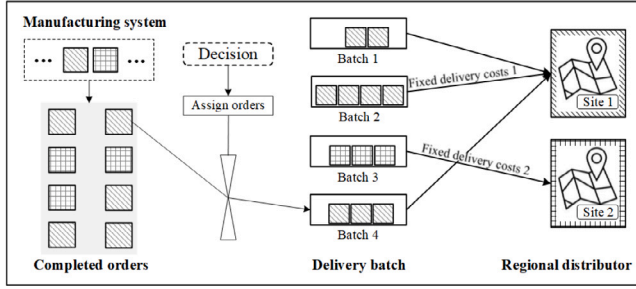
RCFAFSP-BDD combines the scheduling of production orders with the batch delivery to regional distributors. Consequently, it involves two key tasks: first, scheduling production tasks to the appropriate machines during the production stage; and second, assigning completed orders to the suitable delivery batches in the distribution stage.

In the production stage, a set of orders I from multiple regional distributors D must be processed through a three-stage manufacturing

system, consisting of one processing stage and two assembly stages. As depicted in Fig. 2a, each stage features a set of unrelated parallel machines, with machine eligibility constraints applied only in stage 1, and sequence-dependent setup times considered. Each order i has a due date RT_i , and contains at least one work order (i.e., at least one dishwasher product needs to be produced). Each product can be decomposed into multiple pre-associated part, subassembly, and product tasks based on its assembly structure, as shown in Fig. 3. For simplicity, these tasks are referred to as 'items'. Note that part and subassembly items are pre-associated with a specific product item based on the assembly structure, meaning that sharing is not allowed. Additionally, hierarchical coupling constraints must be observed: subsequent items can only start once all pre-associated items are completed. Item j requires a single operation on any available machine $k(k \in M_j)$ in its stage, with processing time $PT_{j,k}$ depending on its size and the machine's capacity. Due to the workshop's large size, buffer areas between stages will not be a bottleneck. Furthermore, the stamping process for the shell requires a specific type of mold for fixation. Therefore, when selecting a machine for item j , it is necessary not only to ensure that machine $k(k \in M_j)$ is idle but also to verify that sufficient mold resources $Res_{j,r}$ are available.



(a) Layout for the three-stage manufacturing system.



(b) Principles of the batch direct delivery.

Fig. 2. Working principles of problems under concern.

In the distribution stage, decisions must be made to assign completed orders to appropriate batches for delivery, using a direct delivery method to transport these batches to regional distributors, as illustrated in Fig. 2b. Under this direct delivery method, each batch is delivered separately to a regional distributor by a vehicle. Consequently, each batch can only include orders for a single regional distributor. The delivery time e_b for batch b is determined by the latest completion time of the orders within the batch. The delivery cost DC_d for each batch is fixed and depends solely on the delivery address d . The total delivery cost is influenced by the number of batches required. If product j is completed early but not delivered on time, inventory costs $IC_{i,j}$ are incurred. Conversely, tardiness costs TC_i are incurred if order i is delayed.

To address the RCFAFSP-BDD, the following tasks must be completed: (i) assign each operation required by an item to an appropriate machine, (ii) sequence the operations on each machine, (iii) determine the start and completion times for each machining operation needed for the items, and (iv) group finished orders into batches. All these tasks must be carried out while adhering to the specific domain features and constraints. The performance measure for this work is to minimize the total cost, which includes inventory costs, delay costs, and delivery costs. Several assumptions underpin this analysis.

- All the orders, machines, and molds are available at time zero.
- Each machine or mold can perform only one item at a time.
- Each item can be processed at exactly one machine.
- Each finished order must be assigned to exactly one batch for delivery to the corresponding regional distributor.
- Transportation time of parts and subassemblies between the three stages is ignored.
- Each batch has no capacity limitation, and sufficient vehicles are provided for delivery.
- No interruption occurs during the process of production and delivery.

3.2. Mixed-integer linear programming model

As a novel optimization problem, there is currently no existing literature on the mathematical model for this problem. The added complexity of allocating auxiliary resources has made developing the mathematical model particularly challenging. The required notation definitions and the MIP model are detailed in Appendices A and B, respectively. These components are designed to serve as benchmarks for validating our proposed CP model.

3.3. Illustrative example

To illustrate and clarify this problem, a simplified example is provided. This example aims to give a general overview of both the production and delivery stages. It features two regional distributors and three orders, with all relevant information detailed in Table 2. Most columns are self-explanatory: the 'Delivery cost' column lists the cost per batch for each distributor, the 'Due date' and 'Tardiness cost' columns specify the due dates, as well as the tardiness costs per unit time for each order, and the 'Inventory cost' column shows the cost per unit time for inventory. The 'Item' column reflects task items derived from the decomposition process shown in Fig. 3 for the product in Fig. 4 (e.g., items 1, 2, and 3 are part tasks for product 1, item 19 is a subassembly task for product 1, and item 24 is a product task for product 1). The 'Mold' column indicates the type of mold required for each item (with only one mold of each type available), and the final column lists the processing time for each item on each machine.

The sample problem is solved using IBM ILOG Constraint Programming Optimizer (version 12.9), which yields an optimal solution with a total cost of 68. Fig. 5 illustrates the order production and batch distribution scheme for this optimal solution, showing that all constraints are met. For instance, hierarchical coupling constraints are respected, such that item 28 starts only after items 18 and 23 are completed. Additionally, auxiliary resource constraints are adhered to, with items 1 and 3 scheduled without overlap in the scheduling horizon because they require the same type of mold, among other considerations.

As depicted in Fig. 5, the delivery times for batch 1, batch 2, and batch 3 are 24, 30, and 50 units of time, respectively. Consequently, both Product 2 and Product 4 incur 4 units of inventory time each. It is important to note that this schedule does not result in any delayed orders. The total cost is therefore calculated as follows:

$$\text{Total cost} = (4 \times 1 + 0 \times 1 + 0 \times 1 + 4 \times 1 + 0 \times 1) + (0 \times 2 + 0 \times 2 + 0 \times 2) + (20 + 20 + 20) = 68$$

4. Constraint programming model

As an NP-hard problem, the MIP model described in Section 3.2 faced challenges in finding high-quality solutions within a reasonable timeframe. Given the success of CP techniques in combinatorial optimization (Fazel Zarandi et al., 2020), this section focuses on developing efficient solution methods using CP. It begins with an overview of CP, followed by an explanation of the variables and functions used in this work. A detailed description of the CP model for the problem is then provided, concluding with a discussion of redundant constraints and search strategies.

4.1. Overview on CP

CP is a programming language designed to capture and manage relationships and constraints among variables, providing a straightforward approach to solving constraint satisfaction problems. Over the years, CP has gained prominence for its efficiency in addressing real-world problems. Its success is attributed to its ability to find feasible solutions for large-scale optimization issues within a reasonable timeframe. By employing specialized solvers, CP leverages constraints to guide the search for solutions. The core model expresses constraints through decision variables, demonstrating particular effectiveness in job shop scheduling and other complex problems by integrating mathematical concepts (Novas, 2019).

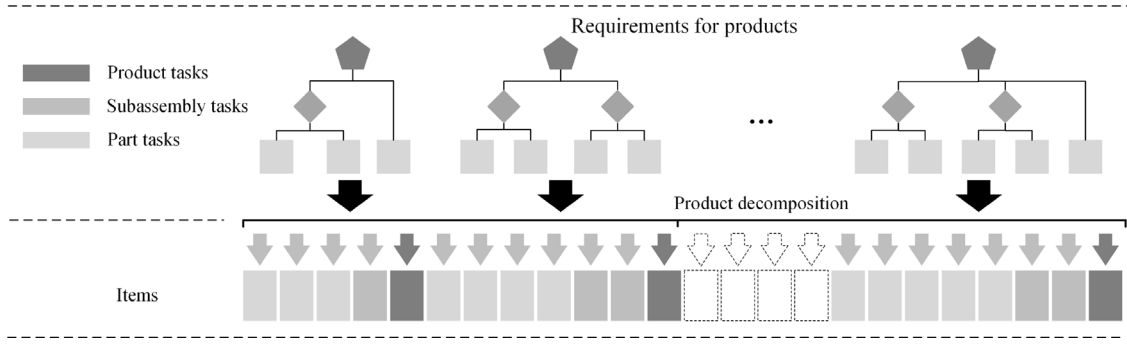


Fig. 3. The process of product decompose.

Table 2

Data for a 2-regional distributor 3-order example.

Regional distributor/No.	Delivery cost/ DC_d	Order/No.	Due date/ RT_i	Tardiness cost/ TC_i	Product/No.	Inventory cost/ $IC_{i,j}$	Item/No.	Mold/ $Res_{j,r}$	Stage1		Stage2	Stage3
									SM1/ $PT_{j,k}$	SM2/ $PT_{j,k}$	PAM/ $PT_{j,k}$	FAM/ $PT_{j,k}$
1	20	1	25	2	1	1	1	2	5	3	—	—
							2	3	8	5	—	—
							3	2	4	7	—	—
							19	—	—	—	5	—
							24	—	—	—	—	4
							4	3	5	3	—	—
		2	30	2	3	1	5	4	5	4	—	—
							6	1	3	4	—	—
							7	1	5	7	—	—
							20	—	—	—	6	—
							25	—	—	—	—	5
							8	2	3	4	—	—
		2	30	2	3	1	9	1	4	3	—	—
							10	1	6	3	—	—
							11	4	4	6	—	—
21	—						—	—	4	—		
26	—						—	—	—	6		
12	3						5	3	—	—		
2	20	3	50	2	4	1	13	2	3	4	—	—
							14	1	3	2	—	—
							22	—	—	—	5	—
							27	—	—	—	—	8
							15	3	5	8	—	—
		5				1	16	1	3	4	—	—
							17	4	5	4	—	—
							18	2	4	3	—	—
							23	—	—	—	7	—
							28	—	—	—	—	4

Abbreviations: SM1/SM2:Stamping machine; PAM:Pre-assembly machine; FAM:Final-assembly machine.

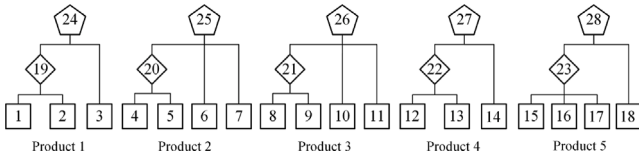


Fig. 4. The assembly structure of the products involved in 2-regional distributor 3-order example.

4.2. Basic principles of using CP solver

In this section, the basic principles of using the CP solver are explained.

4.2.1. Types of variables used in CP solver

1. Interval variable: A time interval during which activities occur, with their exact timing and position not predetermined. These

variables are characterized by their start, end, and size values, which can be retrieved using OPL functions such as *startOf*, *endOf*, and *sizeOf*, as illustrated in Fig. 6. Interval variables can be labeled with the 'optional' keyword, allowing them to be excluded from the solution schedule if desired. Based on this definition, an optional interval variable $X_{j,k}$ is introduced, representing the j th item on the k th machine ($j \in J, k \in M_j$).

2. Interval sequence variable: this variable represents a total order over a set of interval variables. According to this definition, an interval sequence variable Mch_k is defined, which represents the permutation of the present interval variables on the machine k .

4.2.2. Types of functions used in CP solver

1. *startOf*(α): the start time of interval α in the schedule.
2. *endOf*(α): the end time of interval α in the schedule.
3. *sizeOf*(α): the size of interval α in the schedule.
4. *alternative*($\alpha, \{\beta_1 \dots \beta_n\}$): a constraint is utilized to associate a single interval α with one interval from a group of intervals

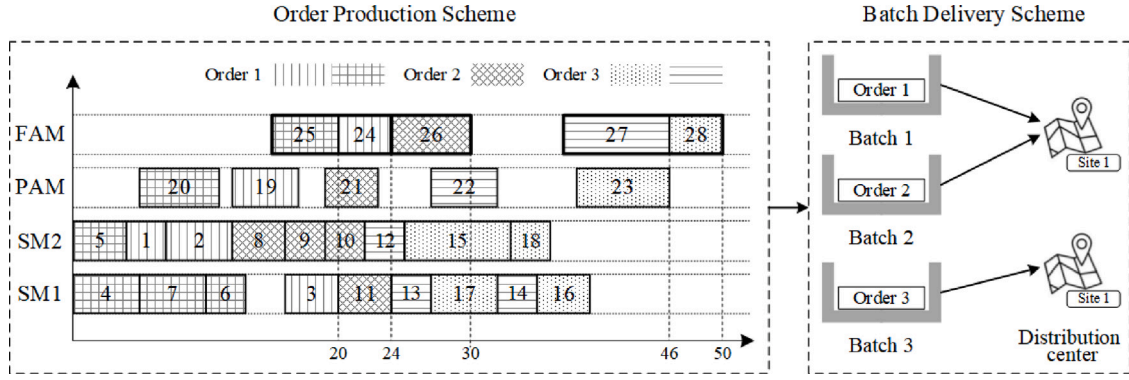


Fig. 5. The schedule and batch delivery result for the mentioned example.

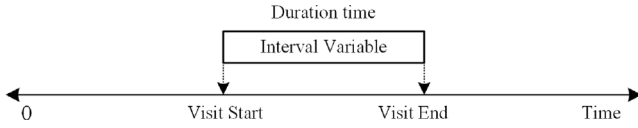


Fig. 6. Representing interval variable.

$\{\beta_1 \dots \beta_n\}$. When interval α is present, only one interval from the set $\{\beta_1 \dots \beta_n\}$ can also be present.

5. *span* (α, β): indicates a constraint that the interval variable α encompasses all intervals in the set $\beta = \{\beta_1 \dots \beta_n\}$. This means that α will start simultaneously with the first present interval in $\{\beta_1 \dots \beta_n\}$ and end with the last present interval in that same set.
6. *presenceOf* (α): which enforces an optional interval α can be either present or absent.
7. *if-Then* (α, β): a rule is employed to execute boolean expression β according to boolean expression α . When boolean expression α is true, then boolean expression β . This function provides a concise and flexible way to perform different operations according to different conditions to satisfy the requirements of the program.
8. *if-Then-Else* (α, β_1, β_2): a rule is utilized to execute a boolean expression β_1 or β_2 based on the boolean expression α . When the boolean expression α is true, the boolean expression β_1 is executed; otherwise, β_2 is executed.
9. *endBeforeStart* (α, β): a constraint which ensures interval α to finish before the start time of interval β .
10. *noOverlap* ($\{\alpha, \beta\}, [T]$): a constraint is essential for scheduling to prevent overlapping intervals in a sequence. It ensures that each interval ends before the next one begins. This constraint becomes particularly significant when there is a transition time between sequential activities, enabling the execution of intermediate tasks.
11. *pulse* (α, h): indicates a cumulative function over the entire scheduling time span, taking value 0 before the start of the interval α and after the end of the interval α ; otherwise, it takes h .

To make it more clear, illustrative graphs are provided in Fig. 7.

4.3. Overview of the developed CP model

This section provides a detailed presentation of the CP model developed for the RCFAFSP-BDD. It begins by introducing the necessary notation, followed by a description of the objective function and constraints.

4.3.1. Notations

The additional notations required for the CP model are given as follows: *Sets*

$M_j = \{1, 2, \dots, |M_j|\}$ Set of eligible machines for each item $j, j \in J, M_j \subseteq M$

$Pr_j = \{1, 2, \dots, |Pr_j|\}$ Set of predecessors of item j , defined by the product's predefined assembly structure, $j \in J, Pr_j \subseteq J$

$Q_i = \{1, 2, \dots, |Q_i|\}$ Set of product items belonging to order $i, i \in I, Q_i \subseteq J$

Decision variables

Z_j Interval variable for item j that must be present in the solution

$X_{j,k}$ Optional interval variable indicating whether item j is assigned to machine k

Mch_k Interval sequence variable for each machine k that includes the interval variables $X_{j,k}$ assigned to machine k

Or_i Interval variable used to span over all product items that belong to order i

Bat_b Optional interval variable used to span over all orders is assigned to b th delivery batch

$Y_{i,b,d}$ Optional interval variable indicating whether order i that belonging to d th regional distributor is assigned to b th delivery batch

$W_{b,d}$ Optional interval variable indicating whether batch b delivered to regional distributor d

4.3.2. Constraints

The constraints to be subjected to are listed as follows:

$$sizeOf(X_{j,k}) = PT_{j,k}, \forall j \in J, k \in M_j \quad (1)$$

This constraint bound the size of interval variables to be equal to the corresponding processing time of items.

$$alternative(Z_j, \{X_{j,k} \mid k \in M_j\}), \forall j \in J \quad (2)$$

Constraint (2) utilizes the CP *alternative* function to ensure that each item is assigned to at most one machine that meets the machine eligibility restrictions. If interval Z_j is present, then exactly one of the intervals $X_{j,k}$ must be present, with Z_j starting and ending with this selected interval.

$$noOverlap(Mch_k, ST_{j,j'}), \forall k \in M \quad (3)$$

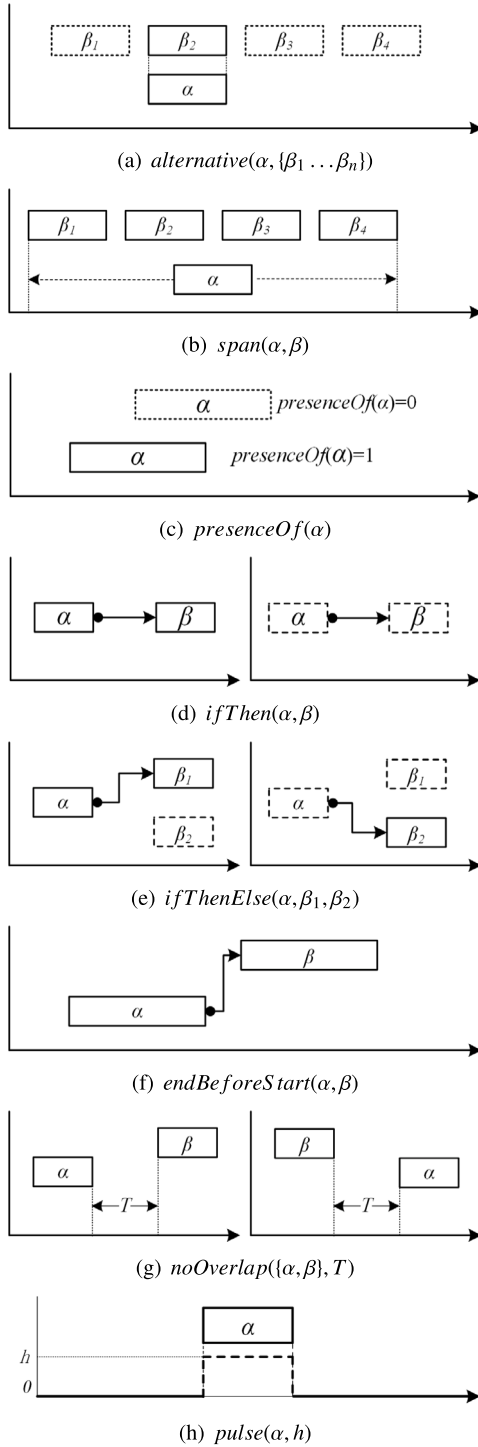


Fig. 7. Illustration for the functions of interval variables: (a) *alternative*; (b) *span*; (c) *presenceOf*; (d) *ifThen*; (e) *ifThenElse*; (f) *endBeforeStart*; (g) *noOverlap*; (h) *pulse*.

Constraint (3) assures a feasible sequence of non-overlapping interval variables $X_{j,k}$ on machine k and embeds setup times $ST_{j,j'}$, between each consecutive time interval pair.

$$endBeforeStart(Z_j, Z_{j'}), \forall j' \in J, j' \parallel Pr_{j'} \mid \neq \emptyset, j \in Pr_{j'} \quad (4)$$

Constraint set (4) is an item precedence constraint set constructed based on hierarchical coupling constraints. For each item pair j' and its immediate predecessor j in the precedence relation set $Pr_{j'}$, the function *endBeforeStart* imposes that item j should be assigned before

item j' .

$$span(Or_i, \{Z_j \mid j \in Q_i\}), \forall i \in I \quad (5)$$

A *span* structure is used to ensure that the interval variable (if present) encompasses all existing interval variables. Constraint (5) is employed to model the requirement that the start and end times of order i must precisely cover the start and end times of a group of product items associated with that order.

$$alternative(Or_i, \{Y_{i,b,d} \mid b \in B, d \in D\}), \forall i \in I \quad (6)$$

Constraint (6) utilizes the *alternative* global constraint to guarantee that each order is assigned to exactly one batch.

$$ifThen\left(\sum_{i \in I} presenceOf(Y_{i,b,d}) > 0, \sum_{i \in I} presenceOf(Y_{i,b,d'}) = 0\right), \forall b \in B, d, d' \in D, d \neq d' \quad (7)$$

Constraint (7) establishes an *ifThen* rule to ensure that a batch can only deliver orders from the same regional distributor. For addressing such problems, employing the *ifThen* rule for designing CP constraints is highly recommended, as this approach shows advantages in both memory usage and solution quality.

$$ifThenElse\left(\sum_{i \in I} presenceOf(Y_{i,b,d}) > 0, presenceOf(W_{b,d}) = 1, presenceOf(W_{b,d}) = 0\right) \forall b \in B, d \in D \quad (8)$$

Constraint (8) ensures that if order i from regional distributor d is assigned to batch b , the interval variable $W_{b,d}$ must be present; otherwise, it must be absent.

$$span(Bat_b, \{Y_{i,b,d} \mid i \in I, d \in D\}), \forall b \in B \quad (9)$$

Constraint (9) is analogous to constraint (5), utilizing the *span* construct. This ensures that delivery batch b encompasses all orders assigned to it, with the delivery time of batch b aligning with the completion time of the last order in that batch.

$$UsageProfile_r = \sum_{j \in J} pulse(Z_j, Res_{j,r}), \forall r \in R \quad (10)$$

$$UsageProfile_r \leq AV_r, \forall r \in R \quad (11)$$

These two constraints are resource capacity constraints that ensure the amount of any resource needed at any point in the scheduling horizon does not exceed the available quantity of that resource.

$$h_{i,j} = \sum_{d \in D} \sum_{b \in B} (endOf(Bat_b) - endOf(Z_j)) \cdot presenceOf(Y_{i,d,b}), \forall i \in I, \forall j \in Q_i \quad (12)$$

Constraint (12) returns the values of inventory time for product j belonging to order i .

$$v_i = \sum_{d \in D} \sum_{b \in B} \max(0, endOf(Bat_b) - RT_i) \cdot presenceOf(Y_{i,d,b}), \forall i \in I \quad (13)$$

Constraint (13) returns the tardiness values for order i .

4.3.3. Objective function

In the present CP model, total cost has been chosen as the performance measure to minimize. Then, expression (14) have to be included in the formulation.

$$\min \sum_{i=1}^I \sum_{j=1}^J h_{i,j} \cdot IC_{i,j} + \sum_{i=1}^I v_i \cdot TC_i + \sum_{d=1}^D \sum_{b=1}^B presenceOf(W_{b,d}) \cdot DC_d \quad (14)$$

Table 3
Workshop environment parameter settings.

Parameters	Values		
	Stamping forming stage	Pre-assembly stage	Final-assembly stage
Number of work orders	Uniform (1, 3)	Uniform (1, 3)	Uniform (1, 3)
Machine flexibility	60%	100%	100%
Processing time of item	Uniform (8, 12)	Uniform (16, 24)	Uniform (34, 46)
Due date of order	Uniform ($L(1 - T - R/2)$, $L(1 - T + R/2)$), $T : 0.2$, $R : 0.8$		
Setup time	4	4	4
Number of item types	6	8	10
Available amount of resource	[3, 2, 2, 1]	–	–
Unit tardiness cost for order	2/order		
Unit inventory cost for product	1/product		
Delivery cost per batch	100/batch		

4.4. Redundant constraints

Constraints (1)–(14) are already effective for finding high-quality solutions for the RCFAFSP-BDD. However, adding redundant constraints is a common technique in CP models that can enhance solution efficiency, even though it is not essential for achieving feasible solutions. Redundant constraints facilitate more comprehensive constraint propagation, leading to stronger reductions in variable domains at each search node and more effective pruning of the search space, ultimately improving search efficiency. Constraints (15)–(16) serve as examples of these beneficial redundant constraints.

$$\text{startOf}(X_{j',k}, L) \geq \max \{ \text{endOf}(X_{j,k'}) \}, \forall j' \in J, j' \parallel Pr_{j'} \neq \emptyset, j \in Pr_{j'}, k \in M_{j'}, k' \in M_j \quad (15)$$

Constraint (15) calculates the earliest possible start time for an item in the second and third stages with the optional interval variable $X_{j',k}$. According to the hierarchical coupling constraints, this start time must be no earlier than the maximum end time of all preceding items $X_{j,k'}$ ($j \in Pr_{j'}$) on eligible machines. To avoid infeasibility when the optional interval variable $X_{j',k}$ is not present, a L value is introduced.

$$\sum_{j \in J} \text{pulse}(Z_j, 1) \leq |M| \quad (16)$$

Constraint (16) ensures that the total machine usage at any point in the scheduling horizon does not exceed the maximum available capacity. When an item is scheduled, the *pulse* function increases the count of occupied machines by one. The cumulative function then restricts this pulse value to ensure it does not surpass the maximum available quantity of machines.

4.5. Proposed search strategy

The CP solver's search phase typically operates with default settings but can be tailored using the OPL language's *searchPhase*. This phase directs the search process through variable and value ordering: the former determines the sequence in which variables are instantiated, while the latter involves sorting the assignments for each variable within its value domain. This study proposes effective search strategies to optimize branching variable and value instantiation order. Specifically, variables are selected based on the smallest domain size using the OPL language *selectSmallest(domainSize())*, and values are chosen according to the smallest expected success rate via *selectSmallest(valueSuccessRate())*.

In search phase 1, interval variables $X_{j,k}$ are utilized to allocate each item to a machine at the appropriate stage, taking into account hierarchical coupling constraints and machine eligibility restrictions. Once the machines for item processing are determined, search phase 2 uses Mch_k to establish the sequence of items on each machine, considering sequence-dependent setup times. In search phase 3, by instantiating interval variable Z_j , each item is assigned to a specific machine, and the start and end times for each item are set according to

the fixed schedule for each machine. This ensures that resource capacity constraints and hierarchical coupling requirements are met. Search phase 4 involves instantiating interval variables Or_i to determine the completion times for each order, based on the relationship between orders and items. Subsequently, search phase 5 uses variables $Y_{i,b,d}$ to assign orders to batches, ensuring that all orders within a batch are for the same regional distributor and that all orders are delivered to the appropriate regional distributors. In search phase 6, variables $W_{b,d}$ are employed to identify the regional distributors for each batch. Finally, interval variables Bat_b are instantiated to accurately determine the delivery times for each batch.

In search strategy 1, it is considered more logical to first address the interval variables $X_{j,k}$ before determining the sequence variables Mch_k . To test this approach, search strategy 2 reverses the order by swapping *Phase 1* and *Phase 2*. Conversely, search strategy 3 is derived by swapping the order of *Phase 5* and *Phase 6* within search strategy 1. This extension aims to explore potential improvements and find a more optimal solution.

5. Experimental results

In this section, Section 5.1 presents the experimental dataset, while Section 5.2 details the numerical results and analysis. One of the main contributions of this work is demonstrating the effectiveness of the proposed CP approach in addressing new and unexplored problems. Therefore, this section evaluates the approach's ability to deliver high-quality solutions across a diverse range of instances and offers practical management implications for manufacturing plants. All experiments utilize commercial solvers IBM ILOG CPLEX and IBM ILOG CP Optimizer (version 12.9) for the MIP and CP models, respectively. The analyses were conducted on a laptop with an AMD Ryzen 7 4800H CPU (2.9 GHz) and 8 GB of memory.

5.1. Instance construction

RCFAFSP-BDD is a novel optimization problem, and there is a notable lack of benchmark instances in the literature. To thoroughly evaluate the proposed MIP and CP models, numerical experiments are conducted using a set of randomly generated test instances. Production datasets are collected from the dishwasher factory, encompassing 200 production orders from seven typical working days. Based on these datasets, empirical probability distribution analyses are performed to learn the characteristics of the factory's production data, including the number of orders and the number of products, etc. These data characteristics serve as guidelines for generating the random sets of instances.

For each instance in the experimental datasets, the following information is generated: the number of regional distributors (D), the number of orders (I), the machine layouts (M), the number of products (P), and the number of items (N). Each order is randomly assigned to a regional distributor, with the number of work orders per order following a uniform distribution $U[1, 3]$. Likewise, each work order

Table 4
Parameter values for generating the instances.

Parameters	Values		
	Small-sized	Medium-sized	Large-sized
Number of orders (I)	2	7, 8, 9	10, 11, 12, 14, 18
Number of products (P)	3, 4, 5	10, 11	16, 18, 30, 31, 32
Number of items (N)	16, 20, 24, 28, 32, 36, 40	60, 80, 100	100, 130, 150, 180, 220, 260
Number of regional distributors (D)	1	3	4, 5
Machine layout ($M1$ - $M2$ - $M3$)	2-1-1	4-3-2	4-3-2, 5-4-3

Table 5
Specifications of the simple case A, B and C.

Case A			Case B			Case C		
Delivery cost per batch	Unit tardiness cost	Unit inventory cost	Delivery cost per batch	Unit tardiness cost	Unit inventory cost	Delivery cost per batch	Unit tardiness cost	Unit inventory cost
100	2	1	20	100	1	20	2	50

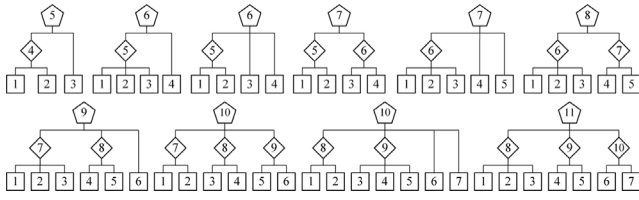


Fig. 8. Assembly structures of the products.

is randomly mapped to a product assembly structure, as illustrated in Fig. 8, involving the production of ten distinct products. Since setup time is influenced by the types of adjacent items, each item is randomly assigned a type. Note that setup time is mandatory for the first item on each machine. Given that the processing of items in the first stage requires auxiliary operations with mold resources, each item is also assigned a mold type requirement at random. Approximately 60% of all alternative machines can perform operations on the items during the first stage. For assembly operations in the second and third stages, any of the alternative machines can be selected, reflecting machine flexibility ($f\%$). Moreover, instance parameters such as processing time ($PT_{j,k}$) for item j on machine k (where $k \subseteq M_j$) and due dates (RT_i) for orders i are generated randomly based on uniform distributions outlined in Table 3. The calculation formula for the L value is provided in Eq. (17). For further details, please refer to the work of Basir et al. (2018). Additionally, all necessary information, including setup time ($ST_{j,j'}$) for each stage, the number of item types per stage, mold resource types, and available quantities for each type (AV_r), as well as unit tardiness costs, unit inventory costs, and unit delivery costs, are fixed at the values specified in Table 3.

Based on the characteristics of the data (I , P , N , D) and the machine layouts ($M1$ - $M2$ - $M3$), the experimental datasets are categorized into small-scale, medium-scale, and large-scale instances, with the feature information summarized in Table 4. The naming convention for these instances follows the format: $|I| \cdot |P| \cdot |N| \cdot |D|$, which indicates the scale of the problem. For instance, '2-3-16-1' represents an instance with 16 items belonging to 2 orders, involving 3 products, and to be delivered to 1 regional distributor. Additionally, for the machine layout, '4-3-2' signifies 4 parallel machines in the first stage, while the second and third stages have 3 and 2 machines, respectively. These instances are designed to be adequately challenging and empirically difficult.

$$L = \max \left\{ \max_{k=1,2,\dots,M-2} \left\{ \sum_{j \in J} PT_{j,k} \right\} + \min_j \left\{ \sum_{i \in I} \sum_{k \in M} PT_{j,k} \times P_{i,j} \right\}, \right. \\ \left. \sum_{i \in I} \sum_{j \in J} \sum_{k \in M} PT_{j,k} \times P_{i,j} \right\} \quad (17)$$

5.2. Results and comparison

5.2.1. Sensitivity testing of CP model to objectives

In this section, the accuracy of the proposed CP model is validated using an example solved by the IBM CP Optimizer. The details of the example, which involves 2 regional distributors and 3 orders, are presented in Table 2. Please note that the unit delivery cost, unit tardiness cost, and unit inventory cost parameters listed in Table 2 will be replaced by those in Table 5 based on specific circumstances.

Three specific cases are considered to test the behavior of the proposed model.

Case A: Assuming that the delivery cost is significantly greater than the tardiness and inventory costs, it is clear that under this assumption it is expected to be possible to have fewer batches. Specifically, all orders from the same regional distributor will be consolidated into a single batch. In this example, the optimal solution is logically derived and compared with the results from the CP Optimizer. The findings are summarized in Table 6.

Case B: Assuming that the tardiness cost is significantly higher than both the delivery and inventory costs, it becomes evident that to mitigate the substantial tardiness costs, a greater number of batches are required. This approach, in turn, results in reduced inventory levels. The results are presented in Table 7.

Case C: Assuming that the inventory cost is significantly higher than both the delivery and tardiness costs, it is clear that to minimize the inventory cost of the product, completed orders will be delivered immediately to the regional distributor. Consequently, this leads to an increased number of batches and further reduces the likelihood of delays. The results are shown in Table 8.

The results from the three cases examined confirm the validity of the mathematical model's behavior. Notably, Tables 6 and 7 reveal that tardiness cost and inventory cost do not exhibit a strict negative correlation. Specifically, it appears that as the quality of tardiness cost in a solution improves, the inventory cost also increases to some extent, and vice versa.

5.2.2. Benefit of integrated production and distribution decisions

In traditional manufacturing, the production and logistics departments have typically operated independently in their decision-making processes as upstream and downstream entities. However, research has highlighted their interdependence, indicating the need for a more holistic approach. Economically, two decision-making modes are considered: (1) separate decision-making within the production and logistics departments, and (2) integrated decision-making across both departments. To compare these two modes, the example presented in Table 2 is solved using the IBM CP Optimizer, and the numerical results for both modes are provided in Table 9.

Table 9 clearly shows that the total cost in the first mode is higher than in the second mode. This suggests that the first mode tends to

Table 6

Results of experiments on the simple case A.

Best lower bound (Obtained logically)	Result of CP optimizer	Number of batches	Number of tardy orders	Number of inventory products	CPU time (s)
217	223	2	1	3	12.32

Table 7

Results of experiments on the simple case B.

Best lower bound (Obtained logically)	Result of CP optimizer	Number of batches	Number of tardy orders	Number of inventory products	CPU time (s)
68	68	3	0	2	8.63

Table 8

Results of experiments on the simple case C.

Best lower bound (Obtained logically)	Result of CP optimizer	Number of batches	Number of tardy orders	Number of inventory products	CPU time (s)
260	268	3	1	1	9.28

Table 9

Results of integrated and non-integrated production and distribution decisions.

	Optimum total cost	Delivery cost	Tardiness cost	Inventory cost
Separate decision-making mode	85	40	26	19
Integrated decision-making mode	68	60	0	8

Table 10

Experiment results on small-size instances.

No.	Instance size	Machine layout	MIP formulation		CP formulation		GAP_{MIP}^{CP}
	$ I \cdot P \cdot N \cdot D $	$M1 \cdot M2 \cdot M3$	Exact	CPU time ^a (s)	Best	CPU time ^a (s)	
1	2-3-16-1	2-1-1	128 ^b	11	128 ^b	0.12	0.00
2	2-3-20-1	2-1-1	132 ^b	115	132 ^b	0.32	0.00
3	2-3-24-1	2-1-1	116 ^b	435	116 ^b	0.83	0.00
4	2-3-28-1	2-1-1	[117, 133] 12%	7200	128	0.64	3.90%
5	2-4-24-1	2-1-1	[135, 137] 1.46%	7200	135	0.44	1.48%
6	2-4-28-1	2-1-1	[119, 139] 13.1%	7200	119	0.41	15.13%
7	2-4-32-1	2-1-1	[138, 185] 25.4%	7200	152	0.69	21.71%
8	2-4-36-1	2-1-1	[128, 216] 40.7%	7200	142	7.87	52.11%
9	2-5-28-1	2-1-1	[163, 175] 6.86%	7200	163	4.01	7.36%
10	2-5-32-1	2-1-1	[165, 200] 17.5%	7200	167	24.05	19.76%
11	2-5-36-1	2-1-1	[135, 155] 12.9%	7200	135	1.15	14.81%
12	2-5-40-1	2-1-1	[137, 235] 41.7%	7200	145	1.32	62.07%
Avg.				5446.8		3.5	

 $|I|$, $|P|$, $|N|$, $|D|$: # orders, # products, # items, # regional distributors.^a Initial time required to instantiate the best solution within 7200 CPU seconds.^b Indicates the optimal solution.

overlook certain effective parameters that could reduce costs in the latter. This comparison demonstrates that integrated optimization of production and distribution problems at a more global level is both more competitive and promising.

5.2.3. Comparison between CP model and MIP model

In this section, due to the NP-hard nature of the problem, it is expected that the MIP approach may encounter challenges. Therefore, we focus on a set of 12 small-sized instances to evaluate the performance of both MIP and CP. To ensure the generalizability of our findings, comparisons are made using the default settings of the solvers. We do not attempt to influence the search parameters, as this is a common practice among researchers comparing MIP and CP technologies. Each model is allocated a maximum CPU time of 7200 s. Given that the CP model produces superior solution values, the gap between the two models is calculated using Eq. (18).

$$GAP_{MIP}^{CP} = \frac{MIP_{sol} - CP_{sol}}{CP_{sol}} \times 100 \quad (18)$$

Most columns in Table 10 are self-explanatory. When optimality is not proven, the ‘Exact’ column displays the best lower bound, best upper bound, and the gap. The data in Table 10 clearly indicate

that the CP model outperformed the MIP model. In instances No. 1–3, both solvers found the optimal solution; however, the CP model demonstrated significantly greater solving efficiency, outperforming the MIP model by one or two orders of magnitude. In the remaining instances, the MIP model failed to prove optimality due to the high number of variables and constraints leading to excessive memory usage, preventing it from finding the optimal solution within the allotted CPU time. The efficient representation of variables and constraints in the CP formulation enables the CP Optimizer to discover high-quality solutions quickly, suggesting that CP may hold greater potential for practical engineering problems compared to the traditional MIP model.

5.2.4. Validity of the proposed redundant constraints

In this section, we further analyze the impact of redundant constraints on the model, as described in Section 4.4. The experimental results for medium-sized and large-sized instances are presented in Tables 11 and 12, respectively. Since the MIP model could not capture a feasible solution within the time limit, only solutions given by the CP model without the feasible cuts and the CP model with the feasible cuts are provided.

Table 11 shows that, compared to CP1, the CP3 model achieves the best values for 13 medium-sized instances and improves the objective

Table 11

Experiment analysis for medium-sized instances.

No.	Instance size I - P - N - D	Machine layout <i>M1-M2-M3</i>	CP1 ^b		CP2 ^c		CP3 ^d		CP4 ^e	
			Best	Time ^a (s)	Best	Time ^a (s)	Best	Time ^a (s)	Best	Time ^a (s)
1	7-10-60-3	4-3-2	942	55.75	940	62.56	904 ^f	48.57	948	89.62
2	7-10-80-3	4-3-2	614	162.62	605	523.52	600 ^f	1564.10	600 ^f	562.90
3	7-10-100-3	4-3-2	696	1277.50	685	1625.23	684 ^f	1427.37	692	1895.62
4	7-11-60-3	4-3-2	1252 ^f	44.84	1259	51.25	1252 ^f	62.79	1262	102.3
5	7-11-80-3	4-3-2	640 ^f	163.50	640 ^f	102.25	640 ^f	89.60	645	158.62
6	7-11-100-3	4-3-2	744 ^f	2314.25	758	1952.36	770	2365.29	768	1652.84
7	8-10-60-3	4-3-2	862	140.09	849	256.25	843 ^f	385.62	864	300.56
8	8-10-80-3	4-3-2	688 ^f	1108.06	698	1325.36	688 ^f	998.23	696	889.6
9	8-10-100-3	4-3-2	605 ^f	2065.34	616	2536.12	622	2848.60	608	2232.5
10	8-11-60-3	4-3-2	1353	721.35	1300 ^f	152.36	1300 ^f	56.81	1352	562.39
11	8-11-80-3	4-3-2	857 ^f	1522.54	857 ^f	358.63	857 ^f	302.32	861	852.6
12	8-11-100-3	4-3-2	800 ^f	1983.87	818	723.54	802	750.52	803	1852.36
13	9-10-60-3	4-3-2	1139	3127.93	1138 ^f	2165.32	1188	1190.90	1153	498.67
14	9-10-80-3	4-3-2	1175	320.67	1049	956.74	1033 ^f	3083.95	1098	3452.21
15	9-10-100-3	4-3-2	686	1687.32	678 ^f	1265.95	678 ^f	1039.06	689	1855.64
16	9-11-60-3	4-3-2	1421	2513.10	1397	563.25	1391 ^f	90.12	1462	1785.66
17	9-11-80-3	4-3-2	985 ^f	1652.79	987	569.61	993	1041.62	989	2334.95
18	9-11-100-3	4-3-2	977	900.18	922	1152.30	832 ^f	413.65	969	1542.57
Avg.				1209		908		897		1257

^a Initial time required to instantiate the best solution within 7200 CPU seconds.^b CP model without cuts.^c CP model with cuts for constraint (15).^d CP model with cuts for constraint (16).^e CP model with cuts for constraints (15)–(16).^f Indicates the optimal solution.**Table 12**

Experiment analysis for large-size instances.

No.	Instance size I - P - N - D	Machine layout <i>M1-M2-M3</i>	CP1 ^b		CP2 ^c		CP3 ^d		CP4 ^e	
			Best	Time (s) ^a	Best	Time (s) ^a	Best	Time (s) ^a	Best	Time (s) ^a
1	10-16-100-4	4-3-2	1539	358.50	1522	328.52	1499 ^f	166.40	1503	686.32
2	10-16-130-4	4-3-2	1257	2766.14	1226	2548.80	1219 ^f	2293.45	1219 ^f	1965.32
3	10-16-150-4	4-3-2	1443	3551.04	1448	685.63	1352 ^f	976.51	1358	1235.24
4	10-18-100-4	4-3-2	1506 ^f	1138.46	1711	546.76	1626	84.22	1755	845.32
5	10-18-130-4	4-3-2	1277	3092.73	1276	986.34	1268 ^f	2415.26	1271	1651.20
6	10-18-150-4	4-3-2	1086	494.18	1093	1845.6	1060	2777.14	1058 ^f	3152.39
7	11-16-100-4	4-3-2	1521 ^f	2770.60	1521 ^f	1253.50	1528	600.23	1522	958.63
8	11-16-130-4	4-3-2	1211	2740.04	1176	986.35	1148 ^f	3040.01	1159	2863.32
9	11-16-150-4	4-3-2	952	845.96	944	2593.60	915 ^f	1281.79	915 ^f	1856.32
10	11-18-100-4	4-3-2	1843 ^f	655.36	1849	1563.55	1875	959.70	1853	2153.22
11	11-18-130-4	4-3-2	1435	3113.14	1412	756.32	1398 ^f	2114.67	1415	2156.62
12	11-18-150-4	4-3-2	1056	2427.42	1065	481.36	975 ^f	3171.00	1002	1756.20
13	12-16-100-4	4-3-2	1604	182.56	1542	754.95	1505 ^f	2304.75	1516	1454.32
14	12-16-130-4	4-3-2	1238	2225.50	1078	3152.60	1076	1598.01	1070 ^f	2565.32
15	12-16-150-4	4-3-2	1110	2113.07	1052 ^f	963.64	1052 ^f	2344.39	1052 ^f	1121.63
16	12-18-100-4	4-3-2	1539	2283.68	1463	3562.51	1424 ^f	760.89	1431	1956.22
17	12-18-130-4	4-3-2	1232	2679.46	1256	1753.21	1203 ^f	1975.46	1213	856.15
18	12-18-150-4	4-3-2	1226	1263.50	1244	751.32	1223	2847.32	1223 ^f	2443.52
19	14-30-180-5	4-3-2	2062	2390.89	1989	562.32	1928 ^f	3541.23	2023	956.32
20	14-30-220-5	4-3-2	1520 ^f	2608.95	1532	1747.60	1730	3304.76	1736	3125.32
21	14-30-260-5	4-3-2	1436 ^f	1825.14	1465	2565.32	1436 ^f	1893.76	1441	1145.23
22	14-31-180-5	4-3-2	2464	3370.15	2419	496.38	2354	2723.21	2349 ^f	979.60
23	14-31-220-5	4-3-2	1480 ^f	2924.03	1480 ^f	3454.96	1559	2020.25	1691	3556.21
24	14-31-260-5	4-3-2	1113 ^f	3270.23	1251	1845.56	1113 ^f	3151.03	1250	2765.32
25	14-32-180-5	4-3-2	2615	2197.43	2601	1989.65	2500 ^f	3088.35	2533	1563.39
26	14-32-220-5	4-3-2	2628	3160.78	2594	1895.63	2587 ^f	3397.96	2587 ^f	3125.32
27	14-32-260-5	4-3-2	2072	2419.84	2208	1562.30	1969 ^f	1431.89	2001	2969.30
28	18-30-180-5	4-3-2	1947 ^f	3054.81	2031	2665.32	2220	3394.93	1947 ^f	2754.41
29	18-30-220-5	4-3-2	1727	2232.32	1720	1786.65	1668 ^f	3189.56	1679	2636.34
30	18-30-260-5	4-3-2	2307	3196.15	2298	1976.32	2201 ^f	3596.76	2432	2982.65
31	18-31-180-5	4-3-2	3257	2758.18	3175	3196.35	3066 ^f	3413.67	3066 ^f	3156.21
32	18-31-220-5	4-3-2	1847	3522.71	1852	3296.54	1902	3345.95	1841 ^f	3591.25
33	18-31-260-5	4-3-2	2147	2856.73	2198	3195.63	2122 ^f	3449.92	2139	3352.28
34	18-32-180-5	4-3-2	2494 ^f	3292.82	2533	3259.83	2594	2390.85	2562	2969.43
35	18-32-220-5	4-3-2	2490	1249.85	2474	2764.52	2406 ^f	2840.20	2406	2559.60

(continued on next page)

Table 12 (continued).

No.	Instance size I - P - N - D	Machine layout M1-M2-M3	CP1 ^b		CP2 ^c		CP3 ^d		CP4 ^e	
			Best	Time (s) ^a	Best	Time (s) ^a	Best	Time (s) ^a	Best	Time (s) ^a
36	18-32-260-5	4-3-2	1917 ^f	3290.12	2236	2984.93	2015	1875.12	2011	2838.45
Avg.				2420		1854		2327		2186

^a Initial time required to instantiate the best solution within 7200 CPU seconds.

^b CP model without cuts.

^c CP model with cuts for constraint (15).

^d CP model with cuts for constraint (16).

^e CP model with cuts for constraints (15)–(16).

^f Indicates the optimal solution.

function for 8 of them. Notably, CP3 demonstrates a 34.78% reduction in average computation time relative to CP1. Specifically, for instance No. 18, CP3 improves the solution value by 17.43% while reducing CPU time by an impressive 117.32% compared to CP1. Although the CP2 model also shows a significant decrease in average CPU time, it is less effective in improving the solution quality. In contrast, CP4 performs worse than CP1, with a 3.97% increase in average computation time. Overall, both CP2 and CP4 do not meet expectations, whereas the CP3 model, utilizing the feasible cut set for constraint (16), shows greater promise. This may be attributed to the large number of redundant constraints introduced by constraint (15), which are not effectively leveraged in medium-sized instances, thereby limiting their advantages.

For the 36 large-sized instances presented in Table 12, the performance of CP3 is notably stronger. Specifically, CP3 achieves the best results in 23 instances, while CP1, CP2, and CP4 secure the best results in only 10, 3, and 11 instances, respectively. CP3 improves upon the solutions obtained by CP1 in 25 instances, though it only reduces the average CPU time by 3.99%. Remarkably, for instance No. 3, CP3 not only improves the solution by 6.73% compared to CP1 but also achieves a dramatic CPU time reduction of 263.83%. While CP4 struggled with medium-sized instances, it shows more promise in large-sized instances, achieving a 10.70% reduction in average CPU time. Although CP2 demonstrates the best performance regarding average CPU time, it does not improve upon the solutions provided by CP1. In conclusion, while the theoretical addition of redundant constraints to CP models can enhance search efficiency, this experiment suggests that such additions should be applied selectively rather than uniformly across all problems.

5.2.5. Effectiveness of the proposed search strategies

In this section, to verify the effectiveness of the proposed search strategies, large-scale instances are employed and tested based on the CP3 model. The numerical results are presented in Table 13.

The data in Table 13 shows that search strategy 1 outperforms or matches the results of CP3 in 25 out of 36 instances, indicating that its design is both scientific and promising. In contrast, search strategy 2 and search strategy 3 only achieve equal or better results than CP3 in 11 and 12 instances, respectively, highlighting their inferiority compared to both search strategy 1 and CP3. This suggests that while customizing a search strategy can theoretically enhance the efficiency of a CP model, developing an effective strategy requires extensive experience and is often a trial-and-error process that can be time-consuming. Notably, the default settings of CP Optimizer for search strategies already demonstrate considerable effectiveness.

5.2.6. Application in a dishwasher factory

To enhance production efficiency and market responsiveness in a real-world dishwasher factory, the proposed CP model is implemented for integrated decision-making in production and distribution. Survey results indicate that planners prioritize order production based on due dates, with earlier due dates receiving precedence. This approach stems from managers' greater concern over the reputational impact of delayed orders compared to potential cost losses from delivery and inventory. Given the complexity of the case studies, comprehensive data can be found in Tables S1 and S2 at github.com/MrZeng-cyber/ROCFZ. A

detailed comparison between the factory's current strategy and the CP model is provided in Table 14.

From Table 14, it is evident that the proposed CP model significantly enhances both efficiency and accuracy compared to the current practices, where planners depend on manual scheduling strategies. Once customer orders are received by the workshop decision system, the CP model can be directly implemented to generate optimal production orders and batch delivery schemes in practice.

5.3. Managerial implications

To facilitate future implementations of the proposed CP model, this article concludes with the managerial implications that can be derived from this study.

- In practice, some optimization objectives may not be strictly mutually exclusive and can even exhibit positive correlations. For objectives that are positively correlated, it is advisable to reasonably reduce the number of objectives, as this can effectively lower the difficulty of the problem and improve the overall optimization process.
- Compared to the traditional MIP model, the CP formulation is clearly more competitive and promising for solving engineering optimization problems, offering significant advantages in both solution efficiency and accuracy.
- Despite the extra time and effort needed to consolidate production and logistics departments, it is evident that collaborative multi-sectoral decision-making in the supply chain can result in cost reductions and enhanced efficiency.

In summary, the results emphasize the importance of integrating optimization decisions from a broader perspective by combining the production and logistics departments, rather than making separate decisions within each department.

6. Conclusion

This work focuses on an integrated scheduling problem in a flexible assembly flow shop with batch delivery, aiming to minimize the total costs associated with inventory, tardiness, and delivery. It incorporates operational constraints such as auxiliary resource restrictions, hierarchical coupling constraints, machine eligibility restrictions, and sequence-dependent setup times. The main contribution of this article is the introduction of the CP model for this problem, alongside a MIP model for comparison. Several redundant constraints and search strategies were developed to enhance the accuracy and efficiency of the solution. A range of instances of varying scales was employed to evaluate both models. The results indicate that the proposed redundant constraints and search strategies significantly improve the performance of the CP model, which successfully generates high-quality feasible schedules within the time limit. This research ultimately aims to achieve overall optimization in enhancing the operational performance of production and distribution processes.

Future research should build upon this study's contributions. For instance, this work assumes that part items and subassembly items are

Table 13
Comparison of the search strategies for large-size problems.

No.	Instance size	Machine layout	CP3		Search strategy (1)		Search strategy (2)		Search strategy (3)	
	$ I \cdot P \cdot N \cdot D $	$M1 \cdot M2 \cdot M3$	Best	Time ^a (s)	Best	Time ^a (s)	Best	Time ^a (s)	Best	Time ^a (s)
1	10-16-100-4	5-4-3	1499	166.40	1592	1247.25	1623	965.63	1542	1521.13
2	10-16-130-4	5-4-3	1219	2293.45	1202 ^b	2760.17	1232	2151.62	1202 ^b	1956.52
3	10-16-150-4	5-4-3	1352	976.51	1489	2574.09	1416	1526.69	1388	3223.62
4	10-18-100-4	5-4-3	1626	84.22	1486 ^b	2510.40	1602 ^b	652.63	1499 ^b	1884.34
5	10-18-130-4	5-4-3	1268	2415.26	1268 ^b	917.03	1331	1654.32	1312	2125.53
6	10-18-150-4	5-4-3	1060	2777.14	1002 ^b	3136.17	1012 ^b	3251.30	1135	2964.32
7	11-16-100-4	5-4-3	1528	600.23	1578	1223.32	1499 ^b	2854.15	1512 ^b	1288.50
8	11-16-130-4	5-4-3	1148	3040.01	1218	3026.85	1196	2851.26	1237	2726.00
9	11-16-150-4	5-4-3	915	1281.79	954	2414.01	909 ^b	1526.30	1021	3524.41
10	11-18-100-4	5-4-3	1875	959.70	1827 ^b	3370.25	1961	1029.52	1911	1774.20
11	11-18-130-4	5-4-3	1398	2114.67	1391 ^b	1857.43	1558	965.45	1391 ^b	1286.65
12	11-18-150-4	5-4-3	975	3171.00	1087	2751.01	1024	3253.64	975 ^b	2896.60
13	12-16-100-4	5-4-3	1505	2304.75	1505 ^b	1590.28	1621	1854.62	1589	2133.56
14	12-16-130-4	5-4-3	1076	1598.01	1076 ^b	3301.06	1081	3212.85	1136	1933.54
15	12-16-150-4	5-4-3	1052	2344.39	922 ^b	3127.62	967 ^b	2653.39	922 ^b	2955.21
16	12-18-100-4	5-4-3	1424	760.89	1530	1010.98	1399 ^b	1594.24	1518	1273.47
17	12-18-130-4	5-4-3	1203	1975.46	1274	1686.28	1232	2129.36	1203 ^b	1559.20
18	12-18-150-4	5-4-3	1223	2847.32	1223 ^b	660.03s	1244	1958.26	1366	863.31
19	14-30-180-5	5-4-3	1928	3541.23	1928 ^b	1138.46	1942	872.63	1928 ^b	1443.53
20	14-30-220-5	5-4-3	1730	3304.76	1730 ^b	1636.92	1826	3522.71	1766	1967.72
21	14-30-260-5	5-4-3	1436	1893.76	1565	2196.68	1436 ^b	2043.92	1498	2465.67
22	14-31-180-5	5-4-3	2354	2723.21	2346 ^b	2142.09	2388	2449.30	2346 ^b	662.18
23	14-31-220-5	5-4-3	1559	2020.25	1559 ^b	1889.21	1712	937.21	1589	3442.65
24	14-31-260-5	5-4-3	1113	3151.03	1113 ^b	3350.37	1126	3022.91	1228	3456.60
25	14-32-180-5	5-4-3	2500	3088.35	2487 ^b	2788.42	2586	3219.60	2487 ^b	2996.31
26	14-32-220-5	5-4-3	2587	3397.96	2587 ^b	1332.98	2587 ^b	1566.90	2621	3451.17
27	14-32-260-5	5-4-3	1969	1431.89	2337	2380.18	2283	2854.42	2133	896.32
28	18-30-180-5	5-4-3	2220	3394.93	2009 ^b	3007.90	2026 ^b	2965.32	2009 ^b	3521.30
29	18-30-220-5	5-4-3	1668	3189.56	1668 ^b	3549.21	1925	2386.32	1714	2855.60
30	18-30-260-5	5-4-3	2201	3596.76	2196 ^b	2906.56	2196 ^b	3182.30	2231	1855.21
31	18-31-180-5	5-4-3	3066	3413.67	3066 ^b	3564.87	3241	3121.96	3174	3364.28
32	18-31-220-5	5-4-3	1902	3345.95	2006	3599.65	1932	3449.62	1918	2862.36
33	18-31-260-5	5-4-3	2122	3449.92	2120 ^b	2344.60	2374	2965.32	2149	956.30
34	18-32-180-5	5-4-3	2594	2390.85	2585 ^b	3219.04	2592 ^b	2854.32	2621	3125.68
35	18-32-220-5	5-4-3	2406	2840.20	2406 ^b	2973.81	2463	2915.21	2533	3521.26
36	18-32-260-5	5-4-3	2015	1875.12	1736 ^b	3581.65	2155	965.32	1736 ^b	2965.32
Avg.				2280		2480		2273		2307

^a Initial time required to instantiate the best solution within 7200 CPU seconds.
^b Indicates an improved solution for CP3.

Table 14
Comparison between factory strategy and CP model.

Case characteristics				Machine layout	Total cost		Ave Imp
$ I $	$ P $	$ N $	$ D $	$M1 \cdot M2 \cdot M3$	CP model	Factory strategy	
10	16	100	4	4 - 3 - 2	1606	2033	26.59%
12	18	150	4	4 - 3 - 2	1252	1682	22.36%

pre-associated with specific products, restricting sharing. While this aligns with prior research, future studies could relax this constraint to allow sharing across multiple products. Additionally, this study consolidates three cost objectives into a single function; future research could utilize the Pareto method to explore the relationships among these objectives more thoroughly. Furthermore, empirical data should be collected to validate the findings derived from the generalized simulation CP model of a resource-constrained flexible assembly flow shop. Lastly, a key challenge for future research is the effective implementation of the CP model in practical settings.

CRedit authorship contribution statement

Chuangfeng Zeng: Writing – review & editing, Writing – original draft, Software, Resources, Project administration, Methodology, Investigation, Data curation, Formal analysis, Validation, Visualization. **Jianjun Liu:** Supervision, Project administration, Investigation, Funding acquisition, Conceptualization, Methodology. **Qinsong Li:** Software, Investigation, Data curation, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Notation (Indices, Sets, Parameters, Decision variables)

The notations required for the MIP model are given as follows:

Indices and sets

$I = \{1, 2, \dots, i, \dots, |I|\}$ Set of orders

$J = \{1, 2, \dots, j, j', j'', \dots, |J|\}$ Set of items

$JP = \{1, 2, \dots, j, \dots, |JP|\}$ Set of items in the first stage

$M = \{1, 2, \dots, k, k', \dots, |M|\}$ Set of machines

$MP = \{1, 2, \dots, k, \dots, |MP|\}$ Set of parallel machines in the first stage

$B = \{1, 2, \dots, b, \dots, |B|\}$ Set of delivery batches

$D = \{1, 2, \dots, d, d', \dots, |D|\}$ Set of regional distributors

$R = \{1, 2, \dots, r, \dots, |R|\}$ Set of mold types

$T = \{1, 2, \dots, t/s, \dots, |T|\}$ Set of time periods in the scheduling horizon

Parameters

$Eg_{j,k}$ Binary parameter, 1 if machine k capable to process item j ; 0, otherwise

$A_{j,j'}$ Binary parameter, 1 if item j and item j' having precedence constraints based on hierarchical coupling constraints; 0, otherwise

$G_{i,d}$ Binary parameter, 1 if order i from regional distributor d ; 0, otherwise

$P_{i,j}$ Binary parameter, 1 if item j belongs to order i ; 0, otherwise

$PT_{j,k}$ Processing time of item j on machine k , $j \in M_j$

$ST_{j,j'}$ Setup time required for switching item j to item j'

RT_i Due date of order i

$IC_{i,j}$ Unit inventory cost for early completion of product item j that belongs to order i

DC_d Delivery cost per batch for regional distributor d

TC_i Unit tardiness penalties for delayed orders i

$Res_{j,r}$ Resource need of item j to mold resource r

AV_r The available amount of mold type r

L A large number

Decision variables

c_j Completion time of item j

f_i Completion time of order i

v_i Delay time of order i

$h_{i,j}$ Inventory time of product item j belongs to order i

e_b Delivery time of the b th batch

$x_{j,j',k}$ Binary variable taking value 1 if item j' is assigned to machine k immediately after item j ; 0, otherwise

$W_{j,t}$ Binary variable taking value 1 if item j ($j \in JP$) finishes at time t in the first stage, and 0 otherwise

$y_{i,d,b}$ Binary variable taking value 1 if order i from d th regional distributor is allocated to b th delivery batch, and 0 otherwise

$Z_{d,b}$ Binary variable taking value 1 if b th batch delivered to regional distributor d , and 0 otherwise

Appendix B. Objective function and constraints

Minimize:

$$\sum_{i \in I} \sum_{j \in J} (IC_{i,j} \times h_{i,j}) + \sum_{b \in B} \sum_{d \in D} (DC_d \times z_{d,b}) + \sum_{i \in I} (TC_i \times v_i) \quad (B.1)$$

Subject to:

$$x_{j,j',k} \leq Eg_{j',k}, \forall j, j' \in J, j \neq j', k \in M \quad (B.2)$$

$$\sum_{j \in J} x_{0,j,k} = 1, \forall k \in M \quad (B.3)$$

$$\sum_{k \in M} \sum_{j=0, j \neq j'}^{J|} x_{j,j',k} = 1, \forall j' \in J \quad (B.4)$$

$$\sum_{j=0, j \neq j'}^{J|} x_{j,j',k} - \sum_{j=0, j \neq j'}^{J|} x_{j',j,k} = 0, \forall j' \in J, k \in M \quad (B.5)$$

$$c_0 = 0 \quad (B.6)$$

$$c_{j'} \geq c_j + PT_{j',k} + ST_{j,j'} + L \times (x_{j,j',k} - 1), \quad \forall j \in J \cup \{0\}, j' \in J, j \neq j', k \in M \quad (B.7)$$

$$c_{j'} \geq A_{j,j'} \times c_j + \sum_{k \in M} \sum_{j''=0, j'' \neq j'}^{J|} (x_{j'',j',k} \times PT_{j',k}), \quad \forall j, j' \in J, j \neq j' \quad (B.8)$$

$$\sum_{t \in T} w_{j,t} = 1, \forall j \in JP \quad (B.9)$$

$$\sum_{t=PT_{j',k}}^{T|} t \times w_{j',t} \geq PT_{j',k} + ST_{j,j'} + \sum_{t=PT_{j,k}}^{T|} t \times w_{j,t} - L \times (1 - x_{j,j',k}), \quad \forall k \in MP, j, j' \in JP, j \neq j' \quad (B.10)$$

$$\sum_{t \in T} t \times w_{j,t} \leq c_j, \forall j \in JP \quad (B.11)$$

$$\sum_{j=0}^{J|} \sum_{j' \in JP, j \neq j'} \sum_{k \in MP}^{t+PT_{j',k}+ST_{j,j'}-1} \sum_{s=t} Res_{j',r} \times w_{j',s} \times x_{j,j',k} \leq AV_r, \quad \forall r \in R, t \in T \quad (B.12)$$

$$f_i + L \times (1 - P_{i,j}) \geq c_j, \forall i \in I, j \in J \quad (B.13)$$

$$\sum_{d \in D} y_{i,d,b} \leq 1, \forall i \in I, b \in B \quad (B.14)$$

$$\sum_{b \in B} y_{i,d,b} = G_{i,d}, \forall i \in I, d \in D \quad (B.15)$$

$$-\sum_{i \in I} y_{i,d,b} + L \times z_{d,b} \geq 0, \forall d \in D, b \in B \quad (B.16)$$

$$\sum_{i \in I} y_{i,d,b} + L \times (1 - z_{d,b}) \geq 0, \forall d \in D, b \in B \quad (B.17)$$

$$e_b + L \times \left(1 - \sum_{d \in D} y_{i,d,b}\right) \geq f_i, \forall i \in I, b \in B \quad (B.18)$$

$$h_{i,j} + L \times \left(1 - \sum_{d \in D} y_{i,d,b}\right) \geq P_{i,j} \times (e_b - c_j), \quad \forall i \in I, j \in J, b \in B \quad (B.19)$$

$$v_i + L \times \left(1 - \sum_{d \in D} y_{i,d,b}\right) \geq e_b - RT_i, \forall i \in I, b \in B \quad (\text{B.20})$$

$$c_j, f_i, e_b, h_{i,j}, v_i \geq 0, \forall i \in I, j \in J, b \in B \quad (\text{B.21})$$

$$x_{j,j',k}, w_{j,t}, y_{i,d,b}, z_{d,b} \in \{0, 1\}, \quad (\text{B.22})$$

$$\forall i \in I, j, j' \in J, k \in M, b \in B, d \in D$$

In the mathematical model, formula (B.1) minimizes the total cost of inventory, tardiness and delivery as the considered objective function. Constraints (B.2) demonstrate machine eligibility. Constraints (B.3) emphasize that the dummy item 0 is placed at the beginning of the sequence before all the items on each machine. Constraints (B.4) guarantee that each item should be processed precisely at one machine. The constraints (B.5) ensure that each item must be preceded by one item and succeeded by another. Constraint (B.6) forces the completion time of dummy item 0 to be equal to zero. Constraints (B.7) guarantee that the completion times c_j and $c_{j'}$ related to item j and item j' are calculated consecutively if item j' is processed immediately after item j on the same machine. Constraints (B.8) impose that the completion time of an item in second stage or third stage is greater than or equal to its assembly time plus the completion time of any one of parts and lower-level subassemblies that has an assembly relationship with it. Constraints (B.9) ensure that each item is completed exactly at one time. Constraints (B.10) and (B.11) indicate that the completion time of an item in a sequence on a machine in the first stage is at least equal to the sum of the completion time of the preceding item, the setup time, and the processing time of the current item. Constraints (B.12) guarantees that for each resource type r , the total amount assigned to items processing at any time is equal to or less than the available amount of that resource type. Constraint (B.13) determines the completion times of an order. Constraint (B.14) ensures that each order is assigned to exactly one batch for delivery. Constraint (B.15) ensures that all orders are delivered to the corresponding regional distributors. Constraints (B.16) and (B.17) enforce the removal of one or more supposed delivery batches that remain empty due to placing several orders within a certain delivery batch. Since a batch is available for delivery only after its last order has been finished, Constraint (B.18) determines the delivery time of a batch at the regional distributor. Constraints (B.19) and (B.20) return the values of inventory time for product item j that belongs to order i and tardy time for order i , respectively. Finally, constraints (B.21) and (B.22) specify the domains of the decision variables.

The proposed model includes non-linear terms in constraint, such as constraint (B.12). However, it is possible to transform it into a tractable linear model. To linearize constraint (B.12), constraints (B.23)–(B.26) are replaced.

$$F_{j,j',k,t} \leq w_{j',t}, \forall j, j' \in JP, j \neq j', k \in MP, t \in T \quad (\text{B.23})$$

$$F_{j,j',k,t} \leq x_{j,j',k}, \forall j, j' \in JP, j \neq j', k \in MP, t \in T \quad (\text{B.24})$$

$$F_{j,j',k,t} \geq x_{j,j',k} + w_{j',t} - 1, \forall j, j' \in JP, j \neq j', k \in MP, t \in T \quad (\text{B.25})$$

$$\sum_{j=0}^{|JP|} \sum_{j' \in JP, j \neq j'} \sum_{k \in MP} \sum_{s=t}^{t+PT_{j',k}+ST_{j,j'}-1} Res_{j',r} \times F_{j,j',k,s} \leq AV_r, \forall r \in R, t \in T \quad (\text{B.26})$$

Now, the resulted model is completely linear in the objective function and the constraints.

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